Evaluation report: default strategy

dataset synthetic-mixed-v1 · 42 cases · embedder hash:v1:1024 · LLM none · vector store local
Generated 2026-10-31 14:59:45 UTC

Headline metrics

Metric	Value	Meaning
Recall@5	90.9%	share of gold passages found in the top 5
Precision@5	18.2%	relevant results in the top 5 (÷5)
MRR	0.909	mean reciprocal rank of the first relevant result
Faithfulness*	88.9%	lexical proxy (not RAGAS)
Answer relevancy*	26.7%	embedding proxy (not RAGAS)
Latency p50 / p95	30 / 43 ms	end-to-end per question
Tokens / query	0	provider-reported
Est. cost / query	$0.00000	from MKA_PRICE_TABLE

Retrieval

MRR 0.909 over 22 cases with gold passages

K	Recall@K	Precision@K	nDCG@K
1	90.9%	90.9%	0.909
3	90.9%	30.3%	0.909
5	90.9%	18.2%	0.909

Answers and citations

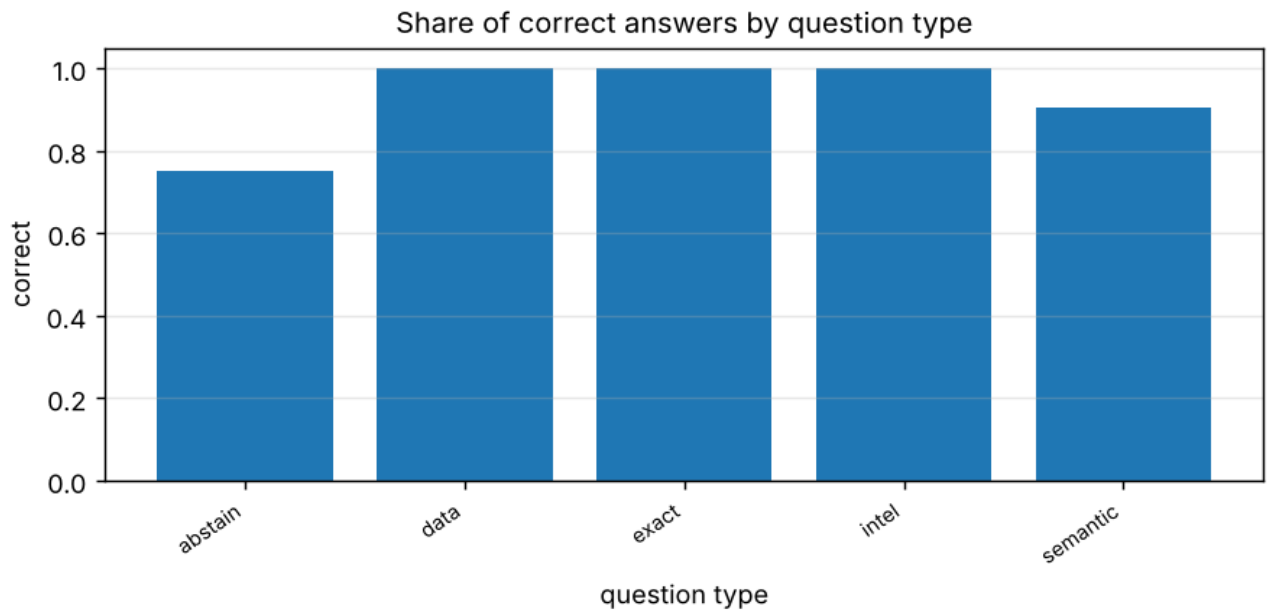
Metric	Value
route_accuracy	100.0%
keyword_accuracy	92.3%
exact_count_accuracy	100.0%
data_accuracy	100.0%
abstention_accuracy	75.0%
abstained_or_flagged_low_confidence	100.0%
answer_rate_on_answerable	97.2%
faithfulness_lexical (proxy)	88.9%
answer_relevancy_embedding (proxy)	26.7%
faithfulness_llm_judge	—
answer_relevancy_llm_judge	—
false_answer_rate_on_unanswerable	25.0%
citation_accuracy	100.0%
citation_gold_precision	60.7%
required_sources_cited	—
stale_or_forbidden_content_absent	—

Agent execution

Measure	Value
Execution success rate	100.0%
Cases with an agent failure	0
Verified answer rate	90.5%
Tool calls	49
Failed tool calls	0
Failed tool call rate	0.0%
Cases with invalid citations	0

By question type

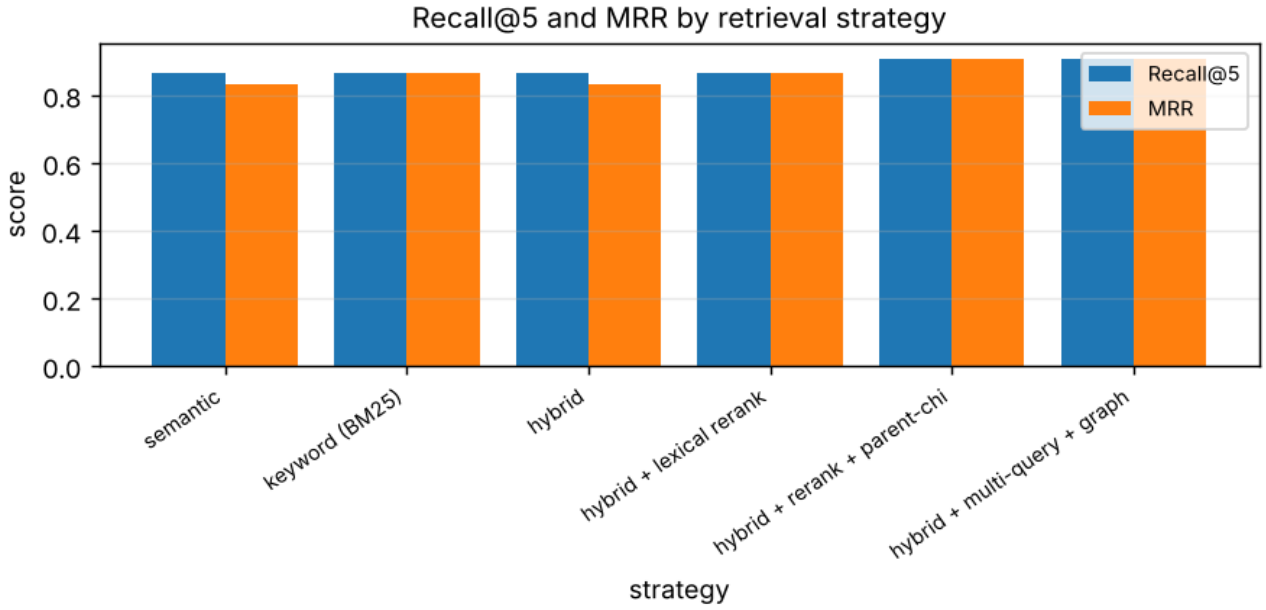
Type	Cases	Correct	Recall@5	MRR	Route accuracy	p50 ms
abstain	4	75.0%	—	—	—	26.4
data	9	100.0%	—	—	100.0%	27.4
exact	6	100.0%	100.0%	1.000	100.0%	19.9
intel	2	100.0%	—	—	100.0%	33.9
semantic	21	90.5%	90.5%	0.905	100.0%	30.4



Strategy comparison

Strategy	Recall@5	Precision@5	MRR	nDCG@5	p50 ms	p95 ms	$/query
semantic	86.4%	17.3%	0.833	0.841	25.5	43.3	0.00000
keyword (BM25)	86.4%	17.3%	0.864	0.864	24.1	31.7	0.00000
hybrid	86.4%	17.3%	0.833	0.841	22.2	37.8	0.00000
hybrid + lexical rerank	86.4%	17.3%	0.864	0.864	26.9	37.1	0.00000
hybrid + rerank + parent-child + corrective	90.9%	18.2%	0.909	0.909	24.8	42.2	0.00000

Strategy	Recall@5	Precision@5	MRR	nDCG@5	p50 ms	p95 ms	$/query
hybrid + multi-query + graph	90.9%	18.2%	0.909	0.909	27.6	40.9	0.00000



Failures (3)

Case	Question	Why
h01	How much vacation time do staff get yearly?	keywords_ok, no relevant passage in top 5
h03	What is the cost of the Pro subscription per month?	keywords_ok, no relevant passage in top 5
a02	What was the weather in Pune yesterday?	abstained_ok

All cases

ID	Type	Route	Verified	Recall@5	ms	Question
s01	semantic	semantic	yes	1.00	79	How many days of annual leave do full-time employees accrue?
s02	semantic	semantic	yes	1.00	39	How many unused leave days can be carried over?
s03	semantic	semantic	yes	1.00	33	When is a doctor's note required for sick leave?
s04	semantic	semantic	yes	1.00	44	How many days a week can employees work remotely?

ID	Type	Route	Verified	Recall@5	ms	Question
s05	semantic	semantic	yes	1.00	40	Within how many days must expense reports be filed?
s06	semantic	semantic	yes	1.00	27	Above what amount are receipts mandatory?
s07	semantic	semantic	yes	1.00	22	How often must passwords be rotated?
s08	semantic	semantic	yes	1.00	28	Within how many hours must a security incident be reported?
s09	semantic	semantic	yes	1.00	30	How long must customer data be retained?
s10	semantic	semantic	yes	1.00	27	What notice is required to terminate the agreement for convenience?
s11	semantic	semantic	yes	1.00	49	Within how many days must invoices be paid?
s12	semantic	semantic	yes	1.00	32	Which law governs the agreement?
s13	semantic	semantic	yes	1.00	40	How long does the confidentiality obligation last after termination?
s14	semantic	semantic	yes	1.00	31	What is the return policy for products?
s15	semantic	semantic	yes	1.00	26	How long is the hardware warranty?
s16	semantic	semantic	yes	1.00	30	When will the mobile app launch?
s17	semantic	semantic	yes	1.00	38	Who signed the reseller agreement with Delta Systems?
s18	semantic	semantic	yes	1.00	30	What is the main risk to the launch?
h01	semantic	semantic	no	0.00	30	How much vacation time do staff get yearly?

ID	Type	Route	Verified	Recall@5	ms	Question
h02	semantic	semantic	yes	1.00	18	How soon do we have to tell the security team about a breach?
h03	semantic	semantic	yes	0.00	30	What is the cost of the Pro subscription per month?
e01	exact	exact	yes	1.00	20	Does Anjali Rao appear in the documents?
e02	exact	exact	yes	—	17	How many times does "Delta Systems" appear?
e03	exact	exact	yes	—	17	Find SKU-1003
e04	exact	exact	yes	—	34	Search for "Zaphod Beeblebrox"
e05	exact	document_search	yes	—	35	Where does clause 8 appear?
e06	exact	exact	yes	—	20	How many times does "Bob Mehta" appear?
d01	data	data	yes	—	30	What is the total sales?
d02	data	data	yes	—	25	Which employee has the highest sales?
d03	data	data	yes	—	26	What is the average sales?
d04	data	data	yes	—	28	What is the total sales in the North region?
d05	data	data	yes	—	27	Which region has the highest total sales?
d06	data	data	yes	—	25	How many customers are in the UK?
d07	data	data	yes	—	29	Are there duplicate emails in customers.csv?
d08	data	data	yes	—	29	Show the top 3 employees sorted by sales
d09	data	data	yes	—	27	What is the median sales?
a01	abstain	semantic	no	—	22	What is the capital of Mongolia?

ID	Type	Route	Verified	Recall@5	ms	Question
a02	abstain	semantic	yes	—	38	What was the weather in Pune yesterday?
a03	abstain	semantic	no	—	31	Who won the football world cup in 1998?
a04	abstain	semantic	no	—	20	What is the CEO's favourite colour?
i01	intel	summarize	yes	—	38	How is Anjali Rao related to Dev Patel?
i02	intel	summarize	yes	—	30	Extract all the entities from meeting_notes.md

Faithfulness/relevancy marked (proxy) are lexical/embedding heuristics, not RAGAS; the LLM-judge values exist only when an LLM is configured.

Token counts are provider-reported when available, otherwise estimated at ~4 characters per token; cost uses MKA_PRICE_TABLE.

citation_accuracy = share of returned citations whose page/row exists and whose excerpt occurs there (checked by the Verification Agent); agent execution success = no internal error or timeout in any agent node.

Sources

Document	Version	Type	Indexed	Hash
hr_policy.pdf	1	pdf	2026-10-31T14:59:43+00:00	ba9b0b05a100
security_policy.pdf	1	pdf	2026-10-31T14:59:43+00:00	dabd44ecf45e
service_agreement.docx	1	docx	2026-10-31T14:59:43+00:00	4c5876822788
sales_q1.xlsx	1	xlsx	2026-10-31T14:59:43+00:00	a3456d50fc4a
customers.csv	1	csv	2026-10-31T14:59:43+00:00	ac028fe25d35
meeting_notes.md	1	md	2026-10-31T14:59:43+00:00	346fd8191637
product_faq.txt	1	txt	2026-10-31T14:59:43+00:00	da8a3daa756b
inventory.json	1	json	2026-10-31T14:59:43+00:00	a85550f8d5d8